Let's combine these variables in a few separate examples. (You can always return to the default prompt with the PROMPT PG command if you make a mistake.) Say you wanted the prompt to show the current date and time along with the drive and location. You would use this command:

PROMPT PGDT

This would alter the prompt to show the current date and time:

C:\>Sun 08/28/201115:08:14.47

However, that prompt looks rather cluttered. It might work better if you put a space between the date and the time with this command:

PROMPT PGSDST

The resultant prompt would look like this:

C:\> Sun 08/28/2011 15:09:17.24

By playing around with the PROMPT command and its variables, you can customize the prompt to your liking.

CHANGING THE TITLE BAR

By default, the title bar of the Command Prompt window reads simply "Command Prompt." You might want to change it to show more information - if you have multiple Command Prompt windows open, changing the title bar is a quick reminder to let you know what task you are performing in each window. Or you could simply change it for your own amusement.

You can change the text of the title bar with the TITLE command. For instance:

TITLE Hello!

This will change the text of the Command Prompt window from "Command Prompt" to "Hello!"

CHANGING THE TEXT AND BACKGROUND COLOR

When you launch a new Command Prompt window, it shows white text on a black background. This a throwback to the old DOS prompt, which always booted up with white text on a black background (or green text on a black background) in order to reduce the chance of the old-style tube monitors burning the text onto the screen. Fortunately, tube monitors have becoming increasingly rare (I don't miss carrying them from building to building) and flat-panel LCD monitors have taken their place.

Reading white text on a black background for a long period of time can put strain on your eyes. If you want to change the color of the Command Prompt's text or background, you can do so with the COLOR command.

The COLOR command generally looks like this:

COLOR 07

If you entered this command at a default Command Prompt window, nothing would happen. Why not? COLOR 07 sets Command Prompt to use white text on a black background. The first digit sets the background color, and the second digit sets the text color. So, COLOR 07 sets Command Prompt to use a black background and white text.

Here is the complete list of codes to use with the COLOR command:

-0, Black.

-1, Blue.

-2, Green.

-3, Aqua.

-4, Red.

-5, Purple.

-6, Yellow.

-7, White.

-8, Gray.

-9, Light Blue.

-A, Light Green.

-B, Light Aqua.

-C, Light Red.

-D, Light Purple.

-E, Light Yellow.

-F, Bright White.

For instance, if you wanted to set the Command Prompt to use white text on a black background, you would use COLOR with these options:

COLOR 70

Or, if you wanted to set Command Prompt to use green text on a black background (to make it look a bit like "The Matrix" science fiction movies), you would use this command:

COLOR 02

Finally, if you make a typo and the Command Prompt is displaying some hideous mixture of colors, like blue text on a black background, simply type the COLOR command without any options. Entering COLOR without any codes after it will reset the prompt back to white text on a black background.

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Chapter 4 - Working with directories

We've already mentioned directories quite a few times in this book. In this chapter, we'll discuss commands for moving from directory to directory, for listing directory contents, creating directories, and deleting directories. With these commands, you will have a thorough knowledge of how to work with directories from the Windows Command Prompt.

CHANGING DIRECTORIES

In the previous chapter, we discussed how the prompt changes as you move from directory to directory. But how exactly do you move from directory to directory in the command line? It's not as if you can click on an icon, after all. Fortunately, the command to change the current directory is quick and easy to type - the CD command, which stands for "change directory." For instance, if Command Prompt is in the root (\) directory of C, and you wanted to change to the Users directory, you would type this command:

CD Users

Once you're in the C:\Users directory, if you wanted to move back up to the root directory, you would use this command:

CD ..

Typing the "CD .." command moves you up one directory level. If you're in C:\Windows\system32, the "CD .." command will move you up to C:\Windows. If you're in C:\Windows, "CD .." will move you up to the root directory.

The CD command will let you move to any subdirectories immediately below the level of the current directory. For instance, if you're in the root directory, you need only type "CD Windows" or "CD Users" to change to the Windows directory or the Users directory.

However, CD will not let you jump to a subdirectory two levels down from your current directory. To do that, you'll need to use the full path of the directory to which you want to move. Let's say you are in the root directory of C, and you want to change to C:\Windows\system32. You could type "CD Windows", and then once you're in C:\Windows, "CD system32." However, it might be quicker just to type this:

CD \Windows\system32

With tab completion, it is definitely quicker to use the full path.

LISTING DIRECTORY CONTENTS

Once you've changed your location to a new directory, the next step is to see the contents of the directory. And you can do that with the DIR command. The DIR command, entered without any switches, displays a list of the files and subdirectories in the current directory. For instance, if you're in the root directory of your C drive, and you type the DIR command, the output will look something like this:

07/13/2009 10:20 PM <DIR> PerfLogs

08/16/2011 05:34 PM <DIR> Program Files

08/12/2011 10:13 PM <DIR> Program Files (x86)

03/08/2011 09:44 PM 6,276 shared.log

08/21/2010 10:43 PM <DIR> SIERRA

04/20/2010 08:00 PM <DIR> Users

11/07/2007 08:00 AM 5,686 vcredist.bmp

11/07/2007 08:50 AM 1,927,956 VC_RED.cab

11/07/2007 08:53 AM 242,176 VC_RED.MSI

07/30/2011 11:08 PM <DIR> Windows

25 File(s) 3,953,909 bytes

6 Dir(s) 811,349,843,968 bytes free

The default output of the DIR command produces five columns of information. The first column displays a list of dates - these dates show the last time the file was modified. The second column shows a list of times. This is called a timestamp - it shows the last time the file was modified, and together with the information of the first column, indicates both the date and the time that the file was last changed. The third column is only used if the file listed is in fact a directory - if you see a <DIR> in the line, that means the line belongs to a subdirectory, not a file. The fourth column is only used if the entry actually belongs to a file - the number is the size (in bytes) of the listed file. (To find its size in megabytes, divide the number by 1024.) The final column lists the names of the files and subdirectories in the current directory.

Without any switches, the DIR command only lists the contents if the current directory. If you're in the root directory of C, but you want to list the contents of C:\Windows\system32, you might use the CD command to change the current directory to C:\Windows\system32, and use the DIR command from there. However, the DIR command supports absolute paths:

DIR C:\Windows\system32

Type that command, and DIR will list the contents of the C:\Windows\system32 directory. If you use the absolute path of any directory with the command, DIR will list the contents of that directory, regardless of the directory you currently occupy, if you have proper NTFS permissions to the directory (we'll discuss permissions more in Chapter 5).

As you have seen, the DIR command presents a variety of useful information. Using command switches, you can alter the information DIR displays and view additional data.

The most common switch used with DIR is the /P switch. Let's say you type this command:

DIR C:\Windows\system32

If you do, the results of the command will quickly scroll out of sight. To prevent this, use the command with a slight alteration:

DIR C:\Windows\system32 /P

Instead of scrolling off the top of the Command Prompt window, DIR will instead display the results one screen at a time. Once you've finished examining a screen of information, press any key to scroll to the next screen of information. Keeping pressing keys until you reach the end of the list and return to the prompt. (Alternatively, you can press CTRL-C to cancel the command and return to the prompt.)

The /A switch is another useful option for the DIR command. Type this command in your home directory:

DIR /A

You'll probably see a lot more files than you usually do.

The /A switch does two things. First, it lets you see any hidden files in a particular directory. When a file is "hidden", it has (as you might expect) the hidden attribute set, which means it doesn't show up in Windows Explorer and doesn't appear in the default output for the DIR command. (We'll discuss file attributes more in Chapter 5.)

Second, you can also use the /A switch to look for files that have specific file attributes enabled. For instance, if you wanted to see only hidden files in the current directory, you would use this command:

DIR /AH

To see only files that have the system attribute set:

DIR /AS

To see only files that have the read-only attribute set:

DIR /AR

To see only directories (this switch will list any subdirectories in the current

directory, but no files):

DIR /AD

It is possible to mix and match the additional letter after the /A switch. For example, if you wanted to view only directories that were hidden:

DIR /AHD

Another useful option for the DIR command is the /X switch. For the default DIR command, the output looks something like this:

11/07/2007 08:00 AM 17,734 eula.1028.txt

11/07/2007 08:00 AM 17,734 eula.1031.txt

11/07/2007 08:00 AM 10,134 eula.1033.txt

11/07/2007 08:00 AM 17,734 eula.1036.txt

11/07/2007 08:00 AM 17,734 eula.1040.txt

11/07/2007 08:00 AM 118 eula.1041.txt

11/07/2007 08:00 AM 17,734 eula.1042.txt

11/07/2007 08:00 AM 17,734 eula.2052.txt

11/07/2007 08:00 AM 17,734 eula.3082.txt

However, if you were to use the DIR /X command in the same directory, the output would instead look like this:

11/07/2007 08:00 AM 17,734 EULA10~1.TXT eula.1028.txt

11/07/2007 08:00 AM 17,734 EULA10~2.TXT eula.1031.txt

11/07/2007 08:00 AM 10,134 EULA10~3.TXT eula.1033.txt

11/07/2007 08:00 AM 17,734 EULA10~4.TXT eula.1036.txt

11/07/2007 08:00 AM 17,734 EUD159~1.TXT eula.1040.txt

11/07/2007 08:00 AM 118 EUFCB1~1.TXT eula.1041.txt

11/07/2007 08:00 AM 17,734 EU37E0~1.TXT eula.1042.txt

11/07/2007 08:00 AM 17,734 EULA20~1.TXT eula.2052.txt

11/07/2007 08:00 AM 17,734 EULA30~1.TXT eula.3082.txt

You'll notice the extra, trunucated file names before the main filename at the end of the row. As we mentioned in Chapter 1, these are "8.3 filenames", shorter versions of the longer file names used in Windows. You can use the DIR /X command to find these shorter 8.3 filenames.

Why would you need to know these? There are a few situations where knowing the shorter filenames would come in handy. Certain kinds of batch files, for instance, can't handle spaces in file names. Knowing the file's 8.3 name would let you use it in a batch file. Furthermore, some older applications cannot handle long file names, in which case you would need to know the 8.3 filenames in order to use the application.

The DIR command generates quite a lot of information, and you might want to sort it for easier readability. You can do this with the /O switch. By itself, DIR /O doesn't do very much - it simply lists any subdirectories alphabetically, and then any files in alphabetical order. Like DIR /A, the usefulness of DIR /O comes in the additional options you can use with /O.

If you wanted to list files by extension, you would use this command:

DIR /OE

This will list the files by type - all the DOC files (Word documents) would be listed together, all the JPEG files, and so forth.

To list the contents of a directory by size, use this command:

DIR /OS

This will list all the files in the directory, from the smallest to the largest.

If you want to list all the files in the directory by date, use this variant of the DIR /O command:

DIR /OD

This will list the files in the currently directory by age, with the oldest at the first of the list and the newest at the bottom of the list.

DIR /OS will list files by size, from smallest to largest, while DIR /OD will list files by age, from oldest to youngest, but what if you wanted to reverse the

order? What if you wanted to list the files from largest to smallest and youngest to oldest? By prefixing a dash (-) to the options of DIR /O, you can reverse the order of the standard listing.

To list files by size, from largest to smallest, use this variant of the DIR command:

DIR /O-S

And to list files by their age, from youngest to oldest, use this variant of the DIR command:

DIR /O-D

Another useful switch with the DIR command is the /Q option. DIR /Q generates the usual output you've come to expect from the command, but with one additional piece of information – it also lists the owner of the files of the directory. In NTFS file permissions (we'll discuss those more in Chapter 5), the owner of a file generally as full control over it, and can assign permissions to other users on the system or network. Using the DIR /Q command, you can find the owner of files from the command line, which is useful when deciding what NTFS permissions to assign (which we'll also discuss in Chapter 5).

The DIR command, by default, only lists the files in the current directory. By modifying the command with the /S switch, DIR will list the contents of the current directory, as well as the contents of any subdirectories – as well as the contents of any subdirectories within those subdirectories. So as you can imagine, DIR /S command often produces quite a considerable amount of

output. In fact, if you use DIR /S in the root directory of your C drive, it will list every single file on your hard drive. Depending upon the speed of your computer and the size of your hard drive, it can sometimes take upward of five minutes to list all the files!

So DIR /S might seem like something of a novelty, but when used in conjunction with the wildcards feature, it is a powerful tool. "Wildcards" are characters that can represent multiple characters, or even all other possible characters. For instance, the question mark character (?) represents one potential character. Let's say you had a series of files in your current directory named Document1.doc, Document2.doc, and so forth up to Document9.doc. To use the question mark wildcard to view just these files, type this command:

DIR Document?.doc

This command will list every single file in the current directory that begins with "Document", has an extension of "doc", and has one additional character before the file extension.

Wildcards are even more useful when you use them with the asterisk (*) wildcard character. The question mark wildcard character only represents a single character. In the example above, the command will list Document1.doc through Document2. doc, but if there's a Document11.doc, the command will miss it, since the question mark wildcard character only represents a single character.

The asterisk, however, represents multiple characters. Let's say you use this command:

DIR *.doc

This command will list every single file with the extension of "doc" in the current directory. Used in combination with wildcard characters, the /S switch suddenly becomes useful. Let's say you wanted to find all the *.doc files on your computer. To do so, you would simply type this command:

DIR *.doc /S

This command will list every single *.doc file on your system. Depending on the size of your hard drive and the number of *.doc files you have, this command might take some time to run. However, you can modify this further with additional combination of wildcards. For example, say you wanted to find every JPEG file on your system that has the word "beach" in the file name:

DIR *beach*.jpeg /S

This command will list every JPEG file on your hard drive that has the word "beach" somewhere in its file name.

The DIR /S command, therefore, can be quite useful for finding files on your hard drive.

CREATING DIRECTORIES

Now that we've discussed how to list the contents of a directory, you might want to create one of your own. To create a directory named "storage", issue this command at the prompt:

MKDIR storage

This will create an empty directory named "storage" in the current directory. Naturally, if you want to create a directory someplace else than the current directory, you'll need to first use the CD command to change your location. Alternatively, the MKDIR command supports using absolute file paths. So if you wanted to create a directory named "temp" in the root directory of your C drive, you would use this command:

MKDIR C:\temp

Note that you can only create directories if you have NTFS permissions of Write or above in the target directory - we'll discuss NTFS permissions in the next chapter.

REMOVING DIRECTORIES

From time to time, a directory will outlive its usefulness, and you'll want to remove it. You can remove directories with the RMDIR command. In its default mode, RMDIR only removes directories if they're empty. Let's say you never used the C:\temp directory from the previous example, and you've decided to get rid of it. To remove the directory, use the RMDIR command:

RMDIR C:\temp

This will remove the "temp" directory.

However, if there are in fact files in the directory, you'll receive this error message:

The directory is not empty.

At this point, you have two options. First, you can enter the directory and manually delete any files and subdirectories inside. Or, you could use the RMDIR command with the /S switch to delete the directory, along with any files and subdirectories it contains. To delete C:\temp and any files or subdirectories it holds:

RMDIR /S C:\temp

Be very careful when using RMDIR with the /S switch! If you're not cautious, you can accidentally wipe out large chunks of your files.

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